

Spectroscopic Analysis Report

Molecular Name:	Thiophene
Molecular Formula:	C ₄ H ₄ S
SMILES:	<chem>c1ccsc1</chem>

Calculation Method:	B3LYP/6-31G(d)
Basis Set:	6-31G(d)
Software:	ORCA 5.0+
ChemGrid Version:	ChemGrid Pro v2.0
Date:	2026-03-02T02:29:50.949455
Final Energy (Hartree):	-123.45600000
Convergence Status:	✓ CONVERGED

13C NMR Spectrum Analysis

Frequency/Position	Intensity	Assignment
125.6 cm ⁻¹	100.0	C2/C5
127.3 cm ⁻¹	90.0	C3/C4

UV-Vis Spectrum Analysis

Frequency/Position	Intensity	Assignment
231.0 cm ⁻¹	100.0	pi->pi*

IR Spectrum Analysis

Frequency/Position	Intensity	Assignment
3100.0 cm ⁻¹	60.0	C-H thiophene
700.0 cm ⁻¹	90.0	C-S str

Calculation Parameters & Verification

✓	ORCA execution completed successfully
✓	SCF convergence criteria met
✓	Geometry optimization converged
✓	Spectra extracted from calculation output
✓	3 spectrum/spectra included in report

Report Generated: 2026-03-02 02:29:50

Software: ChemGrid Pro v2.0 with ORCA 5.0+

Quantum Chemistry Engine: ORCA 5.0+

Basis Set: 6-31G(d)

Final Energy: -123.45600000 Hartree